Exercise 3.3

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$. Classify whether each statement is equivalent to (i) pointwise convergence $f_i \to f$, (ii) uniform convergence $f_i \to f$ uniformly, or (iii) neither.

(a) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i \geq N$:

$$\sup_{x \in A} \|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (ii) uniform convergence. This is exactly the definition via the sup norm.

(b) For all $x \in A$, for all $i \in \mathbb{N}$ there exists $\varepsilon > 0$ such that

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) neither. The statement is always true (take, e.g., $\varepsilon = \|f_i(x) - f(x)\| + 1$), so it does not characterize convergence.

(c) There exists $N \in \mathbb{N}$ such that for all $x \in A$, $i \geq N$, $\varepsilon > 0$:

$$\|f_i(x) - f(x)\| < \varepsilon.$$

Classification: (iii) neither. Since the left-hand side is nonnegative, the inequality for *all* $\varepsilon > 0$ forces $\|f_i(x) - f(x)\| = 0$; i.e. there is N with $f_i \equiv f$ on A for all $i \geq N$. This condition is much stronger than uniform convergence and is not equivalent to it in general.

*d) $\forall \varepsilon > 0\ \forall x \in A\ \exists N \in \mathbb{N}\ \forall i \geq N :\ \|f_i(x) - f(x)\| < \varepsilon.$
Classification: (i) pointwise convergence. The quantifier order matches the definition; N may depend on x and ε.

*e) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A\ \forall i \geq N :\ \|f_i(x) - f(x)\| < \varepsilon.$
Classification: (ii) uniform convergence. Here N works simultaneously for all $x \in A$.

*f) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A :\ (\, i \geq N \ \Leftrightarrow\ \|f_i(x) - f(x)\| < \varepsilon\,).$
Classification: (iii) neither. This is far stronger than uniform convergence (it also demands that for $i < N$ the error is $\geq \varepsilon$ for every x), and it need not hold even when $f_i \to f$ uniformly.

Exercise 3.4

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$ be a sequence of functions. Assume $f_i \to f$ *uniformly* on A. Show that if $B \subset A$, then $f_i \to f$ uniformly on B.

Solution. By uniform convergence on A, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $i \geq N$ and all $x \in A$ we have $\|f_i(x) - f(x)\| < \varepsilon$. Since $B \subset A$, this inequality holds for all $x \in B$ as well, with the *same* N. Hence $f_i \to f$ uniformly on B. Equivalently,

$$\sup_{x \in B} \|f_i(x) - f(x)\| \leq \sup_{x \in A} \|f_i(x) - f(x)\| \xrightarrow{i \to \infty} 0.$$